Typical protein / antibody screening workflow

From DNA constructs to functional evidence: the life science tools behind each step

Example: sequence-defined antibody candidates screened against a protein target

BUILD · Make physical samples | **TEST · Find binders** | **AI learning needs depth per molecule** | **LEARN · Iterate**

Potential constraint: comparable, reproducible assay data

01 / BUILD

Construct design / DNA synthesis

Encode antibody candidates as DNA constructs

OUTPUT
Sequence + construct

REAGENTS
Oligos / gene fragments
PCR enzymes, vectors
Cloning reagents

INSTRUMENTS
Thermocycler
Spectrophotometer
Liquid handling workstation

CONSUMABLES
PCR / deep-well plates
Tubes, pipette tips

REPRESENTATIVE SUPPLIERS
TWIST BIOSCIENCE · GenScript Make Research Easy
Tsingke · Vazyme
Thermo Fisher

02 / BUILD

Protein expression & purification

Produce candidate antibodies and target-protein samples

OUTPUT
Purified samples + yield

REAGENTS
Cells, media, transfection
Affinity resin, buffers

INSTRUMENTS
Shaker / bioreactor*
ÄKTA / FPLC system
Centrifuge

CONSUMABLES
Flasks / culture bags*
Columns, filters, tubes

REPRESENTATIVE SUPPLIERS
cytiva
GenScript · ACROBiosystems
Sino Biological · Gibco
OPM · Duoning

03 / TEST

Primary binding screen

Identify binders and rank initial binding signals

OUTPUT
Hit / binding signal

REAGENTS
Recombinant target proteins
Antibody candidates
Detection reagents, buffers

INSTRUMENTS
ELISA / plate reader
BLI or SPR platform

CONSUMABLES
96 / 384-well plates
BLI sensors / SPR chips
Sealing films, sample tubes

REPRESENTATIVE SUPPLIERS
ACRO BIOSYSTEMS
Sino Biological · Bioss
Sartorius · Cytiva · Tecan
JET Biofil

04 / TEST

Multi-property characterization

Evaluate binding, specificity and developability risks

KD · kon · koff · expression†
Tm · aggregation
polyreactivity · epitope binning

REAGENTS
Reference proteins, dyes
Assay panels, buffers
Analytical reagents

INSTRUMENTS
SPR / BLI; SEC-HPLC
LC / LC-MS
nanoDSF / DLS

CONSUMABLES
Columns, capillaries
Vials, filters, assay plates
Sensors / chips

REPRESENTATIVE SUPPLIERS
NanoMicro
Waters · Agilent · Thermo Fisher
NanoTemper · Sepax
SPR / BLI: Cytiva · Sartorius

05 / TEST

Cell-function validation

Test activity in a relevant cellular model

LABEL
Cell potency

REAGENTS
Cell models, media, cytokines
Staining antibodies
Reporter-assay reagents

INSTRUMENTS
Flow cytometer
High-content imaging
Plate reader

CONSUMABLES
Cell-culture plates / flasks
Pipette tips, filters

REPRESENTATIVE SUPPLIERS
BIOFIL®
Gibco · OPM · Vazyme · Bio-Techne
Molecular Devices · Corning
Waters Biosciences (ex-BD)

06 / LEARN

Data integration & feedback

Link assay results to the next AI design

OUTPUT
Traceable training data

REAGENTS
No direct assay reagents

SYSTEMS / SOFTWARE
LIMS / ELN
Linked data platform
Lab orchestration

CONSUMABLES
No direct assay consumables

REPRESENTATIVE SUPPLIERS
TECAN.
Hamilton
Benchling · Genedata

Sequence + assay context + results → next design cycle

Automation + data orchestration span every step

Wet lab = build + test. AI may test fewer molecules while increasing characterization depth per molecule.

LST demand depends on molecule count × assay depth × net inputs per assay; instrument investment is separate.

Illustrative workflow; assays may run in parallel. * Scale-dependent. † Expression comes from stage 2. Supplier examples and primary sources: SOURCES.md.